

Methodology

STEP 1: Reprocessing the Data

```
chandra_repro indir=. outdir=./repro
```

Why:

This reprocesses raw data with the latest calibrations. Please always do this first to make sure up-to-date corrections (CTI, bad pixels, gain, etc.).

STEP 2: Inspect Events in DS9

```
ds9 repro/acisf<OBSID>_repro_evt2.fits &
```

Why:

Need to visually examine the source to define source and background regions for spectral extraction.

STEP 3: Create Region Files

In DS9:

- Use the “circle” or “ellipse” tool for the source and background.
- Save as source.reg and bkg.reg in FK5 format (sky coordinates).

STEP 4: Extract Spectrum

```
specextract \
infile="repro/acisf<OBSID>_repro_evt2.fits[sky=region(source.reg)]" \
outroot=spec_<OBSID> \
bkgfile="repro/acisf<OBSID>_repro_evt2.fits[sky=region(bkg.reg)]" \
weight=yes \
correctpsf=no
clobber=yes
```

Why:

- This extracts PHA, ARF, and RMF files needed for spectral fitting.
- weight=yes applies area-weighting, important for extended sources.
- correctpsf=no avoids PSF correction (can be yes if source is near chip edge).

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STEP 5: Launch Sherpa for Spectral Fitting

sherpa

Then in Sherpa:

```
EXPLOSION_MJD = 46849.0    # 1987-02-23
NH_FIXED = 0.2             #  $10^{22} \text{ cm}^{-2}$ 
clean()
load_ph("spec_<obsid>.pi", use_errors=True)
ignore(None, 0.5) # ignore energy < 0.5 keV
ignore(8.0, None) # ignore energy > 8.0 keV
```

Why:

We are preparing the data by filtering only the usable energy band and loading response files.

STEP 6: Fit the Spectrum

```
set_stat("cstat")
set_source(xsphabs.abs1 * xsvpshock.plasma)
set_par(abs1.nH, NH_FIXED)
freeze(abs1.nH)
set_par(plasma.kT, 2.0)
set_par(plasma.Tau_u, 1e10)
set_par(plasma.redshift, 0.00093)
```

Ignore reduced statistic and Q-value (C-stat)

Why:

- xsphabs: Absorption by interstellar medium (you fix NH).
- xsvapec: Plasma emission (SN1987A remnant).
- fit(): Sherpa fits model to spectrum.

STEP 7: Plot Energy vs Counts

```
fit()
plot_fit_resid()
```

Orange - model
Blue - energy/counts

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STEP 8: Calculate Absorbed and Unabsorbed Flux

HARD

```
print("Hard absorbed =", hard[0][0])  
print("Hard abs error =", (hard[0][1] - hard[0][2]) / 2)
```

```
print("Hard unabsorbed =", hard[1][0])  
print("Hard unabs error =", (hard[1][1] - hard[1][2]) / 2)
```

SOFT

```
print("Soft absorbed =", soft[0][0])  
print("Soft abs error =", (soft[0][1] - soft[0][2]) / 2)
```

```
print("Soft unabsorbed =", soft[1][0])  
print("Soft unabs error =", (soft[1][1] - soft[1][2]) / 2)
```

Why:

Need both observed and intrinsic fluxes to compare with the literature and track evolution.

STEP 9: Restore NH (VERY IMPORTANT)

```
set_par(abs1.nH, NH_FIXED)  
freeze(abs1.nH)
```

Reason:

Prepares model correctly for next ObsID.

STEP 10: Age calculation

```
MJD_OBS = get_data().header["MJD-OBS"]  
age_days = MJD_OBS - EXPLOSION_MJD  
age_years = age_days / 365.25
```

Reason:

Prepares model correctly for next ObsID.

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STEP 11: Compute age

```
print("Age (days) =", age_days)
print("Age (years) =", age_years)
```

output:

Age of supernova since explosion.

STEP 12: Total counts (after energy filtering)

```
d = get_data()
total_counts = d.counts.sum()
print("Total counts =", total_counts)
```

This respects:

- energy range (0.5–8 keV)
- GTIs
- it loaded PHA

Exposure time:

```
exposure = get_data().exposure
print("Exposure (s) =", exposure)
```

STEP 13: Count-rate + uncertainty

```
import numpy as np

rate = total_counts / exposure
count_err = 1 + np.sqrt(total_counts + 0.75) # Gehrels
rate_err = count_err / exposure

print("Rate (cts/s) =", rate)
print("Rate error  =", rate_err)
```

Optional: Save Fit Results

```
save_model("fit_model_<obsid>.mod")
```

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Model Used

What it is: The spectral model used.

In this case:

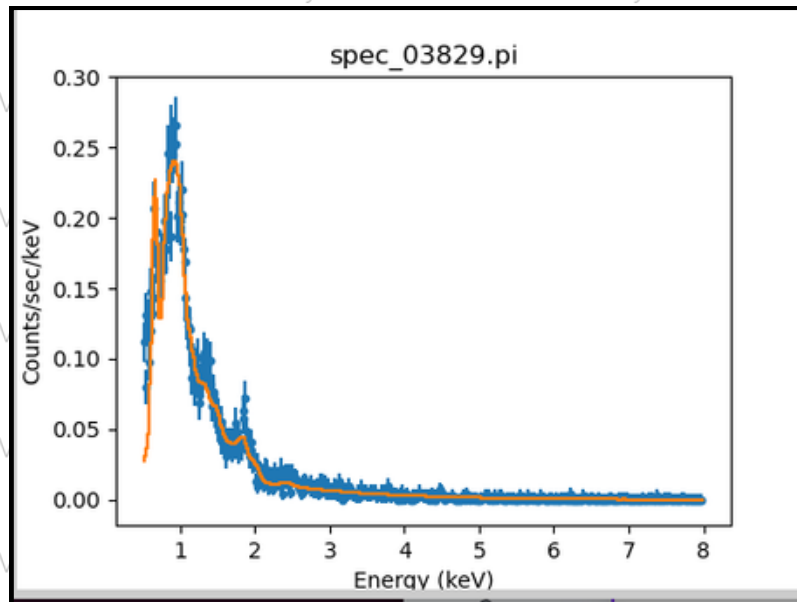
Model: `xsphabs.abs1 * xsvpshock.plasma`

for date

`dmkeypar repro/acisf<obsid>_repro_evt2.fits DATE-OBS echo+`

`dmkeypar repro/acisf<obsid>_repro_evt2.fits TSTART echo+`

`dmkeypar repro/acisf<obsid>_repro_evt2.fits TSTOP echo+`



On the graph, the orange line must lie within the blue region.

Methodology

Step	Goal
chandra_repro	Calibrate and clean raw data
DS9	Visually define regions
specextract	Generate spectral files
Sherpa Setup	Prepare spectrum and models
Fit	Derive temperature (kT), validate model
Flux	Measure brightness (soft/hard, absorbed/unabsorbed)
Excel	Record ObsID, fluxes, temperature, model used